A GenAI-Powered Feedback System for Automated Short-Answer Assessment in Education

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***Abstract*—Automated assessment of short-answer questions remains a significant challenge in educational technology, with traditional approaches struggling to deliver timely, consistent, and pedagogically meaningful feedback at scale. This paper presents a GenAI-powered feedback system leveraging Azure OpenAI's GPT-4o model to provide instant, structured, and actionable feedback for short-answer questions across multiple domains. The system features a modular three-tier architecture, advanced prompt engineering, and domain-specific evaluation strategies. We detail the system methodology, design rationale, iterative development process, and present empirical results and user feedback, demonstrating the effectiveness and scalability of GenAI-powered assessment.**

***Index Terms*—automated assessment, short-answer grading, generative AI, feedback system, educational technology, prompt engineering, large language model (LLM)**

I. Introduction

Timely and constructive feedback is critical for effective learning and student engagement [1]. However, traditional grading of short-answer questions is labor-intensive, slow, and difficult to scale, often resulting in delayed or inconsistent feedback. Recent advances in large language models (LLMs), such as GPT-4o, have enabled new possibilities for automating complex linguistic tasks in education [2]. Despite this, practical systems capable of delivering nuanced, actionable feedback for diverse question types remain underdeveloped.

The integration of artificial intelligence in educational contexts has shown significant promise in enhancing personalized learning experiences [3]. Research demonstrates that AI-driven systems can effectively adapt to individual learning needs and improve educational outcomes across diverse contexts [4]. Furthermore, the emergence of generative AI models has opened new opportunities for automated assessment and feedback generation in higher education settings [5].

The importance of user interface (UI) design in educational AI systems cannot be overstated, as research demonstrates that thoughtful UI design significantly impacts user engagement and learning outcomes [6]. Effective feedback presentation and user-centered design principles are crucial for ensuring that AI-powered educational tools achieve their pedagogical objectives [7, 8].

This work introduces a GenAI-powered feedback system designed to address these challenges, supporting a wide range of question types and providing instant, structured feedback to both students and educators. The system integrates advanced prompt engineering and a modular three-tier architecture to ensure scalability, maintainability, and pedagogical alignment.

II. Related Work

AI-powered educational assessment has evolved significantly, with research highlighting AI's potential to make learning more personalized and accessible [1, 3]. Zawacki- Richter et al. [3] and Bhutoria [4] demonstrate how AI-driven systems can adapt to individual learning needs across diverse cultural contexts, while Naseer et al. [5] show that LLMs like GPT-4o can effectively simulate human-like understanding for grading short-answer questions.

Educational data mining and learning analytics provide crucial foundations for intelligent assessment systems [9], while recent AI tools in academic writing demonstrate growing potential for educational assistance [10]. However, Kasneci et al. [2] emphasize the need for thoughtful integration of AI tools, addressing challenges including academic integrity and pedagogical effectiveness.

User-centered design principles are critical in educational AI systems. Cen et al. [6] demonstrate that thoughtful UI design significantly improves user engagement, while Chang et al. [7] establish key design principles for AI interfaces emphasizing feedback presentation and personalization. Maier and Klotz [8] provide frameworks for structured feedback systems, and Skjuve et al. [11] offer insights into user expectations in AI-powered educational contexts.

Early automated assessment systems relied on keyword matching and statistical methods, lacking semantic understanding. Recent approaches utilize machine learning and natural language processing but often fail to provide context-sensitive, pedagogically relevant feedback. The cognitive foundation of automated feedback systems aligns with Vygotsky's Zone of Proximal Development theory, which emphasizes optimal learning through targeted instruction [12].

Despite these advances, most existing systems focus on binary correctness or require extensive domain-specific customization, limiting scalability. This work addresses these gaps by developing a unified framework capable of delivering detailed, actionable feedback across various question types and domains.

III. Methodology and Implementation

A. Research Method

This project integrated software engineering best practices with state-of-the-art AI techniques through iterative design and development cycles. We systematically tested various prompt engineering strategies and evaluation algorithms to identify approaches yielding pedagogically sound feedback with high accuracy.

A pragmatic research approach combined objective measurement (quantitative system metrics) and subjective interpretation (qualitative user feedback), enabling actionable insights for both technical performance and pedagogical value.

The primary dependent variables included feedback accuracy (measured as percentage alignment with expert human assessment), feedback comprehensiveness (evaluated on a 5-point Likert scale), system response time (measured in seconds per request), and user satisfaction (assessed via post-test questionnaires using a 7-point Likert scale). Independent variables included prompt template structure, question type (factual, explanatory, mathematical, programming), response length, and API temperature settings.

B. System Architecture and Technology Stack

A three-tier architecture (presentation, application, and data layers) was adopted for better maintainability and scalability. This modular separation enables independent testing and easy integration with external services, particularly Microsoft Azure's OpenAI API, while allowing individual components to be improved or replaced without affecting the overall system. React was chosen for the frontend due to its component-based architecture, supporting responsive UI and modularity. The component-based architecture allowed for the modularization of interface elements, ensuring they were reusable and could be independently written and tested. This modularity was especially useful in educational applications where various question formats needed custom components to present questions and answer feedback.

Flask (Python) was selected for the backend, providing lightweight RESTful API support and seamless integration with Azure OpenAI's Python SDK. Using Python with Flask was a strategic choice, as it provided perfect communication between the application server and the LLM service. Flask's lightweight nature also helped the system's performance by adding very little overhead to request processing and response generation. Figure 1 demonstrates the systematic approach to feedback generation and error handling mechanisms. This implementation ensures robust performance and reliable feedback delivery while maintaining system stability, illustrating the complete workflow from user input through the three-tier architecture to final feedback delivery.

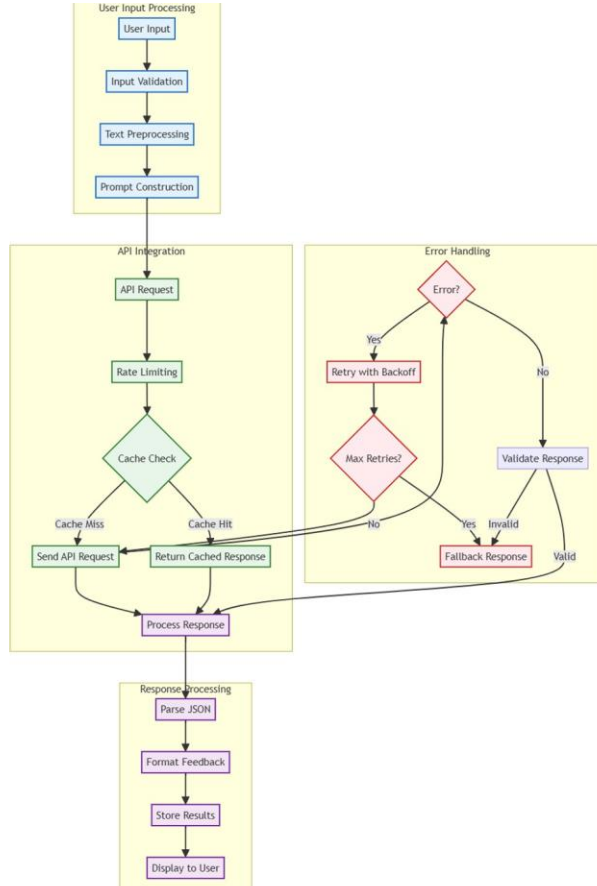

Fig. 1. Technical Implementation Flowchart

C. Prompt Engineering Methodology

This project called for a structured prompt engineering methodology rather than ad-hoc prompt building. This approach was steeped in educational assessment literature and emphasized the importance of consistency and clarity in feedback as central to successful outcomes.

1) Question Type Classification System: The system first determines the question type through sophisticated pattern recognition, categorizing questions into distinct types:

- **Factual questions**: Simple fact-based questions requiring specific answers
- **Explanatory questions**: Questions requiring detailed conceptual explanation
- **STEM questions**: Problems involving science, technology, engineering, and mathematics, including mathematical calculations, chemical formulas, and physical principles
- **Programming questions**: Questions involving code writing, algorithm design, and program logic
- **Analysis questions**: Questions requiring comparison, evaluation, or critical thinking

Figure 2 shows the question classification, template selection, and feedback generation process. This systematic

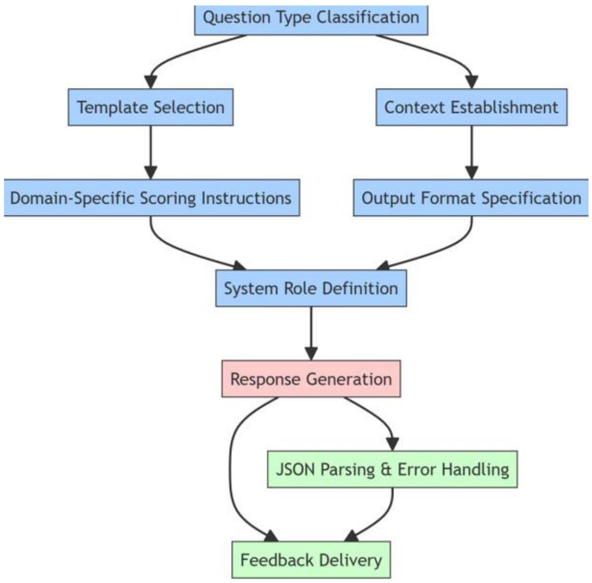

Fig. 2. Prompt Engineering Flowchart

approach ensures consistent and pedagogically sound feedback across different question types, demonstrating how the system automatically selects appropriate assessment strategies based on question characteristics.

2) Template-Based Prompt Architecture: For each question type, we developed specialized prompt templates with four distinct sections:

- **Context Establishment**: Provides the AI with the necessary background information
- **Domain-Specific Scoring Instructions**: Tailored guidelines based on question type
- **Output Format Specification**: Ensures consistent structured feedback
- **System Role Definition**: Sets appropriate AI persona for the assessment task

3) Example Prompt Template for Programming Questions: For programming questions, we developed specialized templates that evaluate multiple dimensions of code quality. The template instructs the AI to assess: (1) correctness of syntax and implementation, (2) algorithm efficiency and complexity, (3) code organization and readability, and (4) proper handling of edge cases.

The scoring rubric provides specific guidelines: answers with a correct approach but minor syntax errors receive 70-85 points, while answers with correct output but inefficient implementation receive 60-75 points. The system generates structured JSON feedback containing a numerical score (0-100), a detailed explanation, identified strengths, suggested improvements, and comprehensive guidance on optimal solutions.

4) Iterative Refinement Process: Our prompt engineering approach incorporated a refinement process through several iterative developmental stages. Initial template development established underlying prompt structures grounded in educational assessment practices. These templates were thoroughly evaluated through test cases using a wide range of sample queries and student responses of varying correctness and complexity.

The feedback generated was analyzed after each round of tests for quality, consistency, and educational value, with emphasis on scoring accuracy, feedback specificity, and pedagogical alignment. Restructured templates were implemented to address identified weaknesses, focusing on strengthening performance patterns and balancing generalizability with domain-specific guidance.

5) Error Handling and Resilience: The system implements advanced JSON parsing with fallback strategies to ensure reliable feedback generation despite varying API responses. The primary method uses plain JSON parsing for faster processing of well-formed responses. If this fails, the system initiates a cascading series of fallback mechanisms, including JSON object extraction, formatting error correction, and structured response creation as a last resort. This layered approach ensures that students receive meaningful feedback regardless of technical issues.

D. Data Collection and Analysis

A reference answer dataset was systematically built to encompass comprehensive answers across various subject domains. Key concepts, expected terminology, and typical misconceptions were carefully annotated in each reference answer to improve the system's ability to generate relevant feedback. The dataset was developed through guided collaboration with domain experts and analysis of authoritative educational resources.

A diverse collection of student responses was compiled to conduct full-scale system testing. These responses had varying levels of correctness, completeness, and conceptual understanding, creating a diverse dataset to test the system's feedback generation capabilities. Domain experts scored these responses to create annotated ground truth references for evaluation.

A comprehensive evaluation framework was developed to analyze feedback quality, prompt engineering effectiveness, and user experience. This multi-dimensional approach integrated technical accuracy assessment with pedagogical efficacy evaluation, supporting the continuous optimization of both prompt engineering strategies and user interface design. Figure 3 outlines the approach used for analyzing system performance and feedback quality. This framework integrates technical accuracy assessment with pedagogical efficacy evaluation, providing a structured methodology for continuous system improvement and validation of educational effectiveness.

The framework included feedback quality assessment (comparing system feedback with human assessments), prompt engineering optimization (analyzing response quality, consistency, efficiency, and adaptability), and user experience analysis (evaluating navigation efficiency, information clarity, task completion, and user satisfaction).

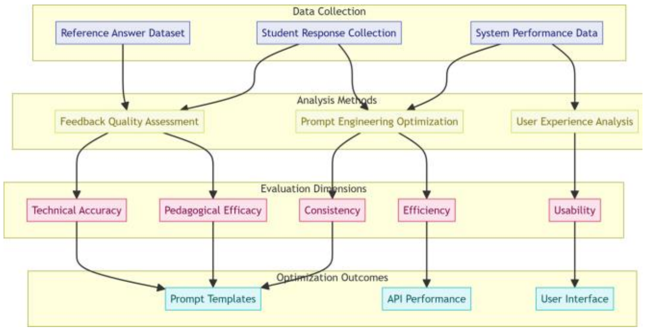

Fig. 3. Comprehensive Evaluation Framework

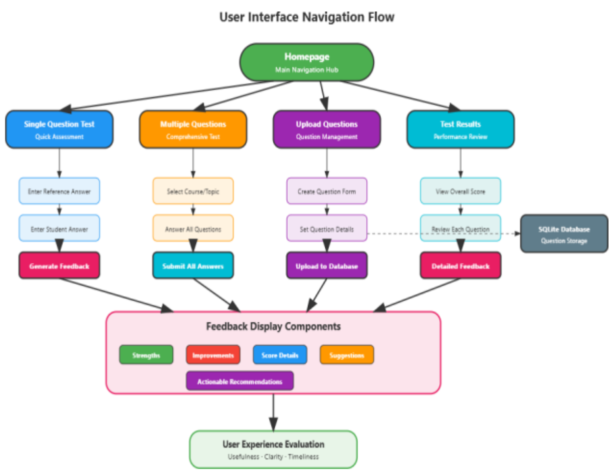

Fig. 4. User Interface Navigation Flow

IV. User Interface Design and Implementation

The user interface was designed with educational usability principles, emphasizing clarity, accessibility, and intuitive navigation. The system implements a clean, professional design that minimizes cognitive load while maximizing functionality, with distinct pathways for single-question assessment, multiple-question testing, and educator management. Figure 4 shows the systematic progression from homepage to assessment completion. This design ensures an intuitive user experience while maintaining pedagogical effectiveness, illustrating the complete user journey from initial access through question interaction to final feedback delivery.

The feedback presentation system employs evidence-based design principles [6–8], using color-coded visual cues to enhance comprehension. Information is presented in digestible sections with a responsive design ensuring optimal functionality across devices, including accessibility features for diverse user needs.

V. Evaluation and Results

A. System Implementation

The system implements a professional user interface with clear navigation options and distinct workflows for single-question assessment, multiple-question assessment, and question management. The interface design emphasizes clarity and accessibility, with color-coded feedback presentation (green for strengths, red for areas for improvement) to enhance readability and comprehension.

The user interface flow supports structured feedback delivery that aligns with the prompt engineering methodology. Users can navigate from the homepage to test pages, interact with different question types, and receive immediate feedback. This design ensures an intuitive experience while maintaining the pedagogical effectiveness of the feedback system.

B. User Study and Feedback Metrics

User feedback was collected from 16 participants, who rated the system on usefulness, clarity, and timeliness. The average usefulness rating was 4.2/5, clarity 4.1/5, and timeliness 4.1/5. Qualitative comments highlighted the clarity and actionable nature of the feedback, with suggestions for more examples and improved handling of alternative answers.

C. Performance Analysis

The system was evaluated through a combination of empirical testing, user feedback, and performance analysis. Key metrics included feedback accuracy, response time, and user satisfaction across diverse question types and subject domains. A total of 34 answers were evaluated across multiple domains. Empirical results indicate that the system delivers consistent, high-quality feedback with rapid response times. Feedback quality analysis reveals that the system successfully identifies key concepts in student responses and provides relevant guidance based on comparison with reference responses.

D. Limitations of the Selected Approach

Although the three-tier architecture and selected technology stack proved fruitful in many areas, implementation discovered some limitations. The ability to use external API services introduced risks and potential dependencies, as system performance was partially reliant on the availability and response times of Azure OpenAI. We addressed this shortcoming by implementing caching and fallback protocols.

The template-based prompt engineering approach offered consistency benefits but did not always provide flexibility for niche question types. This constraint required domain-specific variants for prompting to cater to unique assessment needs for some subjects.

Additionally, the choice of SQLite database, while legitimate for development and testing, could pose limits for large-scale deployment. This limitation was noted in the system design documentation, with recommendations for database migration strategies outlined for future production implementations.

E. Discussion

The GenAI-powered system demonstrates that LLM-based automated assessment can match or approach human-level feedback quality in short-answer questions, with significant gains in speed and scalability. The modular architecture and prompt engineering strategies contribute to robustness, adaptability, and pedagogical alignment.

Several factors contribute to the system's effectiveness:

- **Advanced prompt engineering**: The template-based approach ensures consistent, pedagogically sound feedback across different question types
- **Three-tier architecture**: Clear separation of concerns enhances maintainability and scalability
- **User-centered interface design**: Following established principles for educational AI systems [6], the interface design prioritizes user engagement and learning effectiveness
- **Evidence-based feedback presentation**: The color-coded visual system aligns with research on personalized feedback in digital learning environments [8]
- **Domain-specific evaluation algorithms**: Enable nuanced assessment across various subject areas
- **Responsive user interface**: Ensures accessibility across different devices and user contexts, following established design principles for educational AI interfaces [7]

Limitations include challenges with highly creative or unconventional answers and potential bias towards reference answer formats. The system's performance may vary across disciplines, with stronger results in fact-based domains compared to more interpretive subjects. Additionally, the current implementation has not been thoroughly tested in varied cultural and linguistic settings.

Future research directions include cross-cultural validation studies, advanced prompt engineering techniques for creative responses, integration with learning management systems, and longitudinal impact evaluation of AI-generated feedback on student learning outcomes over time.

VI. Conclusion

This paper presents the methodology and empirical evaluation of a GenAI-powered feedback system for automated short-answer assessment. The system provides a scalable, maintainable, and pedagogically sound solution for enhancing formative assessment in education. User studies and empirical results confirm its effectiveness across diverse domains.

The system's modular architecture, advanced prompt engineering methodology, and robust error handling contribute to its technical success, while its structured feedback approach and intuitive user interface design enhance its educational value. The color-coded presentation of strengths and areas for improvement facilitates student understanding and learning.

The research contributes to the emerging field of AI-informed education by establishing a practical framework for automated short-answer assessment that maintains pedagogical integrity while addressing scalability challenges. The findings indicate that, when thoughtfully implemented, GenAI solutions can effectively complement traditional assessment methodologies, ultimately enhancing the educational experience for both students and educators.

Ongoing research will address current limitations and explore broader applications in educational technology, including multi-language support, deeper integration with learning management systems, and long-term learning outcomes evaluation.

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References

[1] S. Yu and Y. Lu, *An Introduction to Artificial Intelligence in Education*. Singapore: Springer, 2021. [Online]. Available: https://link.springer.com/book/10.1007/978-981-16-2770-5

[2] E. Kasneci, K. Seßler, S. Ku¨chemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Gu¨nnemann, E. Hu¨llermeier, S. Krusche, G. Kutyniok, T. Michaeli, C. Nerdel, J. Pfeffer, A. Poquet, M. Sailer, A. Schmidt, T. Seidel, M. Stadler, J. Weller, J. Kuhn, and G. Kasneci, "ChatGPT for good? On opportunities and challenges of large language models for education," *Learning and Individual Differences*, vol. 103, p. 102274, 2023. [Online]. Available: https://doi.org/10.1016/j.lindif.2023.102274

[3] O. Zawacki-Richter, V. I. Mar´ın, M. Bond, and F. Gouverneur, "Systematic review of research on artificial intelligence applications in higher education—Where are the educators?" *Int. J. Educ. Technol. High. Educ.*, vol. 16, no. 1, pp. 1–27, 2019. [Online]. Available: https://doi.org/10.1186/s41239-019-0171-0

[4] A. Bhutoria, "Personalized education and artificial intelligence in the United States, China, and India: A systematic review using a human-in-the-loop model," *Computers and Education: Artificial Intelligence*, vol. 3, p. 100068, 2022. [Online]. Available: https://doi.org/10.1016/j.caeai.2022.100068

[5] F. Naseer, M. U. Khalid, N. Ayub, A. Rasool, T. Abbas, and M. W. Afzal, "Automated assessment and feedback in higher education using generative AI," in *Transforming Education With Generative AI: Prompt Engineering and Synthetic Content Creation*. Hershey, PA, USA: IGI Global, 2024, pp. 433–461. [Online]. Available: https://doi.org/10.4018/979-8-3693-1351-0.ch021

[6] C. Cen, G. Luo, L. Li, Y. Liang, K. Li, T. Jiang, and Q. Xiong, "User-centered software design: User interface redesign for Blockly–Electron, artificial intelligence educational software for primary and secondary schools," *Sustainability*, vol. 15, no. 6, p. 5232, 2023. [Online]. Available: https://doi.org/10.3390/su15065232

[7] D. H. Chang, M. P.-C. Lin, S. Hajian, and Q. Q. Wang, "Educational design principles of using AI chatbot that supports self-regulated learning in education: Goal setting, feedback, and personalization," *Sustainability*, vol. 15, no. 17, p. 12921, 2023. [Online]. Available: https://doi.org/10.3390/su151712921

[8] U. Maier and C. Klotz, "Personalized feedback in digital learning environments: Classification framework and literature review," *Computers and Education: Artificial Intelligence*, vol. 3, p. 100080, 2022. [Online]. Available: https://doi.org/10.1016/j.caeai.2022.100080

[9] R. S. Baker, T. Martin, and L. M. Rossi, "Educational data mining and learning analytics," in *The Wiley Handbook of Cognition and Assessment: Frameworks, Methodologies, and Applications*. Hoboken, NJ, USA: Wiley, 2016, pp. 379–396. [Online]. Available: https://doi.org/10.1002/9781118956588.ch16

[10] D. Adams and K. M. Chuah, "Artificial intelligence-based tools in research writing: Current trends and future potentials," in *Artificial Intelligence in Higher Education*. Boca Raton, FL, USA: CRC Press, 2022, pp. 169–184. [Online]. Available: https://doi.org/10.1201/9781003184157-9

[11] M. Skjuve, A. Følstad, and P. B. Brandtzaeg, "The user experience of ChatGPT: Findings from a questionnaire study of early users," in *Proc. 2023 CHI Conf. Human Factors in Computing Systems*, Hamburg, Germany, Apr. 2023, pp. 1–18. [Online]. Available: https://doi.org/10.1145/3571884.3597144

[12] C. G. Demartini, L. Sciascia, A. Bosso, and F. Manuri, "Artificial intelligence bringing improvements to adaptive learning in education: A case study," *Sustainability*, vol. 16, no. 3, p. 1347, 2024. [Online]. Available: https://doi.org/10.3390/su16031347